

SecretVR: Differential Privacy Defense Against Membership Inference Privacy Attacks in Virtual Reality

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ABSTRACT

The convergence of artificial intelligence (AI) and virtual reality (VR) technologies (AI VR) offers innovative applications but raises privacy concerns due to the sensitive nature of data (e.g., eye-tracking). Since adversaries could exploit this data to infer and leak sensitive personal information and manipulate user experiences. We demonstrate that AI VR models such as machine learning (ML)/deep learning (DL) models are vulnerable to membership inference attacks (MIA) and leak users' information, with a high success rate. To address this, we propose the SecretVR framework, which uses differential privacy (DP)-enabled privacy-preserving mechanism to defend against MIA and thus protect user privacy in AI VR models. Evaluated on seven AI VR models and three different datasets (i.e., cybersickness, emotion, and activity classification), our approach reduces MIA success rates by up to 32% for VR cybersickness classification tasks while maintaining high model utility, with classification accuracies of up to 92.5% using LSTM AI VR models.

Index Terms: Human-centered computing—Human Computer Interaction—Virtual Reality—Systems evaluation for HCI design

1 INTRODUCTION

Virtual reality (VR) simulation data, specifically eye movement data, is important in human cognition, visual attention, and perception, and it enhances VR experiences by enabling adaptive support, improving engagement, and increasing usability. Furthermore, other eye-related information is commonly utilized in VR with the help of artificial intelligence (AI)-based techniques, such as machine learning (ML) and deep learning (DL), which are widely applied for tasks (e.g., emotion recognition). However, these approaches raise significant privacy and security concerns, as these models train millions of parameters, which can inadvertently memorize and leak sensitive user data, including personal details and training features, and even facilitate re-identification problems. Even anonymized VR datasets remain vulnerable to re-identification attacks (e.g., Netflix Prize challenge). Thus, researchers have applied privacy mechanisms (e.g., differential privacy (DP)) to gaze data. However, these approaches applied privacy mechanisms in the eye-tracking sample levels to make the dataset private while ignoring the ML/DL part. Since eye-tracking data are already noisy sources of information, adding additional noise to preserve privacy may ruin the user experience. Indeed, these ML/DL methods are also vulnerable to different privacy attacks (e.g., membership inference attacks (MIA) [4]). Despite their relevance, the application and defense against MIA in VR remain unexplored, motivating this study.

This paper introduces *SecretVR*, a novel framework providing formal privacy guarantees for ML/DL models in VR applications through DP. Unlike existing methods focusing solely on sample-level privacy, *SecretVR* applies ϵ -DP during AI VR model training,

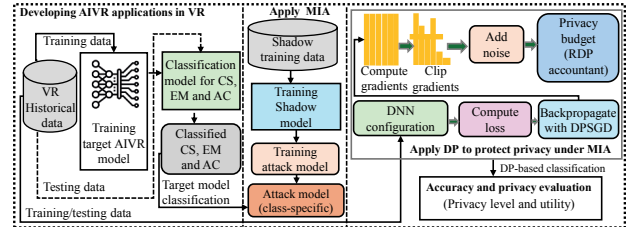


Figure 1: An overview of *SecretVR* framework.

balancing privacy and utility, thus ensuring quantifiable resilience against MIAs. For this, at first, we develop several ML/DL models for three VR applications classifying cybersickness [1], emotions [5], and activity classification [2] of VR users. Then, we apply the MIA against these developed ML/DL models and show that MIA causes serious privacy concerns in VR applications with high success rates of 81%, 76%, and 78% for cybersickness, emotion, and activity classification using LSTM-based classification models, respectively. To defend against MIA attacks and ensure the privacy of AI VR applications, DP mechanisms are employed during training, effectively reducing MIA success rates by up to 32%, 26%, and 27% for cybersickness, emotion, and activity classification tasks using private AI VR LSTM model. Furthermore, from the utility perspective, our results show that the proposed defense has minimal effect on utility with classification accuracies of 92.5% (cybersickness), 89% (emotion), and 93.4% (activity) tasks for the same AIVR model.

2 PROPOSED METHODOLOGY

Figure 1 shows our proposed *SecretVR* framework for privacy-enabled VR prediction to protect user privacy against MIAs. The foremost step in our proposed methodology is to develop ML/DL models for specific VR applications. Our target application is classifying user cybersickness, emotions, and activity in VR. For these three applications, we train three DL models, specifically convolutional neural network (CNN), long short-term memory (LSTM), multilayer perceptron (MLP), and four ML models, namely random forest (RF), decision trees (DT), logistic regression (LR), and Gaussian naive Bayes (GNB). We trained these models using the Simulation 2021 dataset [1], VREED dataset [5], and EHTask dataset [2] for the cybersickness, emotion, and activity classification. Then, we apply MIA against these developed VR ML/DL models. We assume that the primary goal of an attacker is to determine whether a specific individual's data was part of the training dataset used to train an ML/DL model for prediction tasks in VR to leak VR user privacy. To execute the MIA on ML/DL models in VR, we make a similar assumption proposed by Shorki et al. [4]. Thus, the adversary trains N number of shadow models ($N = 20$ for cybersickness, $N = 10$ for emotion, and $N = 15$ for activity) to achieve a higher attack success rate (ASR). Then, the adversary creates and trains an attack model and employs the trained attack model to query the target model and infer membership. Finally, we employ DP to defense against MIAs of these VR ML/DL models. DP is a theoretical definition of privacy that offers a stronger notion of privacy guarantee for output data distribution and computations involving aggregate datasets. To

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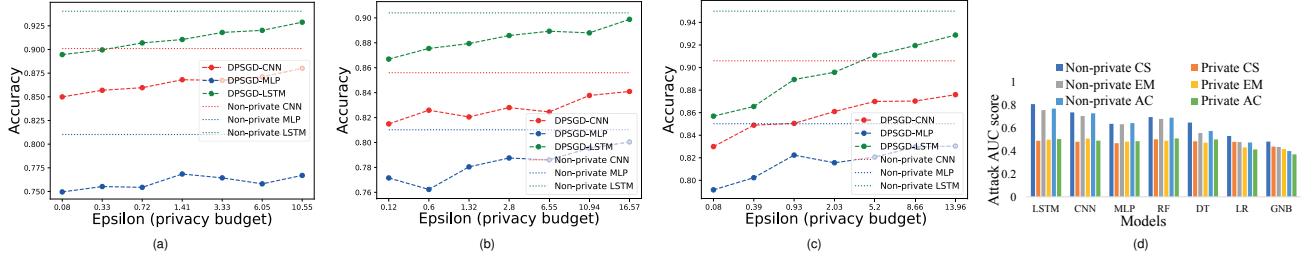


Figure 2: DP-SGD based DL model vs. non-private DL models accuracy for (a) cybersickness (CS), (b) emotion (EM), and (c) activity (AC) classification tasks, and (d) ASR for different private and non-private models under MIA.

Table 1: The ASR of identifying the members in training data regarding balanced accuracy and area under the curve (AUC) score with multiple shadow models for the CS, EM, and AC classification tasks.

Models	Att. balanced acc.			Attack AUC		
	CS	EM	AC	CS	EM	AC
LSTM	0.85	0.80	0.81	0.81	0.76	0.77
CNN	0.75	0.72	0.74	0.74	0.71	0.73
MLP	0.57	0.67	0.69	0.64	0.64	0.70
RF	0.73	0.71	0.70	0.69	0.68	0.70
DT	0.51	0.59	0.57	0.62	0.55	0.58
LR	0.54	0.55	0.51	0.53	0.47	0.48
GNB	0.44	0.49	0.42	0.49	0.44	0.40

implement DP on the ML models, we add noise in two stages during the model training phases, and the parameters of the ML model are generated. In contrast, to protect AI VR DL models against MIAs, our differentially private DL model has two major components: (1) Differentially Private Stochastic Gradient Descent (DP-SGD), which ensures DP during DL model training by incorporating Gaussian additive noise with gradient updates in Stochastic Gradient Descent (SGD) [3] and (2) moments accountant [3] that provide a tighter bound on privacy budget spending. We trained these DL models for m epochs ($m = 200$ for cybersickness, $m = 100$ for emotion, and $m = 130$ for activity classification) at each noise level, followed by calculating the privacy budget and accuracy of the final epoch.

3 EXPERIMENTAL RESULTS

For the cybersickness, emotion, and activity classification task, we train seven AI VR models in a non-private manner and without MIAs. The non-private LSTM, CNN, MLP, RF, DT, LR, and GNB models result in balanced accuracy of 0.93, 0.89, 0.79, 0.87, 0.80, 0.75, and 0.70 for cybersickness severity, 0.90, 0.85, 0.81, 0.86, 0.77, 0.71, and 0.63 for the emotion and 0.95, 0.91, 0.85, 0.84, 0.80, 0.72, and 0.66 for activity classification tasks. We observe that the overall performance of the LSTM model is better than that of the other models for these tasks. Furthermore, Table 1 summarizes the attack balanced accuracy and AUC score under the MIA for the cybersickness, emotion recognition, and activity classification tasks using three DL models and four ML models. We observe that balanced accuracy is higher for the LSTM model, indicating a high ASR. For instance, in the cybersickness classification tasks, the success of an MIA using the LSTM model exhibits an AUC score of 0.82, which is $1.09\times$, $1.26\times$, $1.18\times$, $1.31\times$, $1.54\times$, and $1.68\times$ higher than CNN, MLP, RF, DT, LR, and GNB models. A similar phenomenon is also observed for other classification tasks.

Figure 2(a), (b), and (c) show the mean accuracy of these classification tasks using different DL methods for different privacy budgets ϵ and baseline conditions (without DP) under MIA. Note that the DL-based model performs better than ML-based models in non-private classification and MIA attack evaluation tasks, as shown in Table 1; thus, we evaluate only our defense mechanism against MIA using DP for DL-based classification models. We use the privacy parameters $\delta = 0.00007, 0.00005, \text{ and } 0.00008$ for the

cybersickness, emotion, and activity classification tasks. We observe that private DL models, specifically the LSTM model, achieve higher accuracy in private and non-private settings than other DL models for these classification tasks. We observe that relaxing the privacy budgets ϵ and parameters δ significantly increases the classification task performance of the private models. In contrast, it is observed that there is a considerable accuracy drop for other DL models with low privacy budgets. A similar phenomenon is also observed for emotion and activity classification tasks. We also evaluate how much the private models leak information about their training data. Figure 2(d) shows an ASR of the MIA against different private and non-private models. We observe that the ASR against the corresponding DP-enabled target private models is insignificant and lower than the baseline non-private models. For instance, the private LSTM model reduces the ASR up to 32%, 27%, and 26% than the non-private LSTM model, proving that the DP promises to offer protection against sophisticated attack methods.

4 CONCLUSION

This work proposed a privacy-preserving method to protect user privacy against MIAs for VR emotion recognition, cybersickness, and activity classification AI models. Our experimental results revealed that MIA on AI VR models can significantly leak users' privacy with an 81% ASR on the LSTM AI VR model using the Simulation 2021 dataset. However, applying the DP-enabled defense reduced this rate by 32% while maintaining high utility with up to 92.5% accuracy for the same dataset. A similar phenomenon was observed in other datasets.

5 ACKNOWLEDGEMENT

This material is based upon work supported by the National Science Foundation (NSF) under Award Number CNS-2114035 and partially supported by the Army Research Office (ARO) under award number W911NF-23-1-0401.

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